

AI BASED NETWORK VULNERABILITY ASSESSMENT TOOL

Design Experience

Introduction

- ❑ AI-Based Network Vulnerability Assessment Tool focuses on improving cybersecurity by automatically identifying weaknesses in network systems.
- ❑ With the rapid increase in cyber threats, traditional manual testing methods are becoming slow and inefficient.
- ❑ This project uses Artificial Intelligence to scan networks, detect vulnerabilities, and analyze security risks.

Literature Review

❑ Vulnerability Detection & Assessment

[1] Khan & Parkinson (2018) — Looked at all the ways AI can be used to find security holes in computer systems. They sorted methods from fully manual to fully automatic, and found the hardest part is teaching AI what vulnerabilities look like.

[2] Malkawi & Alhajj (2026) — Studied 29 research papers and showed that AI finds security problems faster and more accurately than old manual methods. But two big problems remain: there is no standard way to test these AI tools, and most AI decisions are hard to understand or explain.

[5] Chopra et al. (2024) — Built a tool called ChatNVD that uses AI (like ChatGPT) to read a huge database of known security bugs and explain them in simple language, so security teams can understand and fix them faster.

[3] Chemparathy et al. (2025) — Asked real cybersecurity professionals about AI-powered scanning tools. Most found them useful, but many still don't use them regularly because they don't fully trust the AI or don't know how to use it well.

Literature Review

❑ AI-Enhanced Intrusion Detection

[6] Susilo et al. (2025) — Built an AI system that watches network traffic on smart devices (IoT) and spots attacks automatically. It uses two AI layers working together — one to understand the data, another to detect suspicious patterns over time. Works well even when attack examples are rare.

[12] Yu et al. (2026) — Used AI to scan billions of internet-connected devices quickly and smartly, instead of checking them one by one. The AI learns which devices to check first based on location and speed, making the scan much faster.

[7] Podder et al. (2025) — Built a system called SPEAR that maps out all the possible paths an attacker could take through a network. It lets security teams ask "what if" questions and get clear advice on how to block those paths.

Overall finding: AI systems that combine multiple techniques always work better than those using just one method. Making AI decisions easy to understand for humans is still a major challenge.

Literature Review

❑ LLM & RL-Driven Automated Pentesting

[8] Deng et al. (2023) — Built PentestGPT, an AI tool that tries to hack into systems the same way a human security tester would. It is split into three parts so it doesn't "forget" what it was doing. It performed 228% better than basic AI at completing hacking tasks.

[9] Patil et al. (2024) — Created a framework where AI automatically does the full security test — gathering info, finding weak points, trying to break in, and writing a report — using real-world attack patterns as a guide.

[11] Yu et al. (2025) — Trained an AI agent to figure out the best route through a network to carry out an attack, helping security teams understand where their biggest risks are.

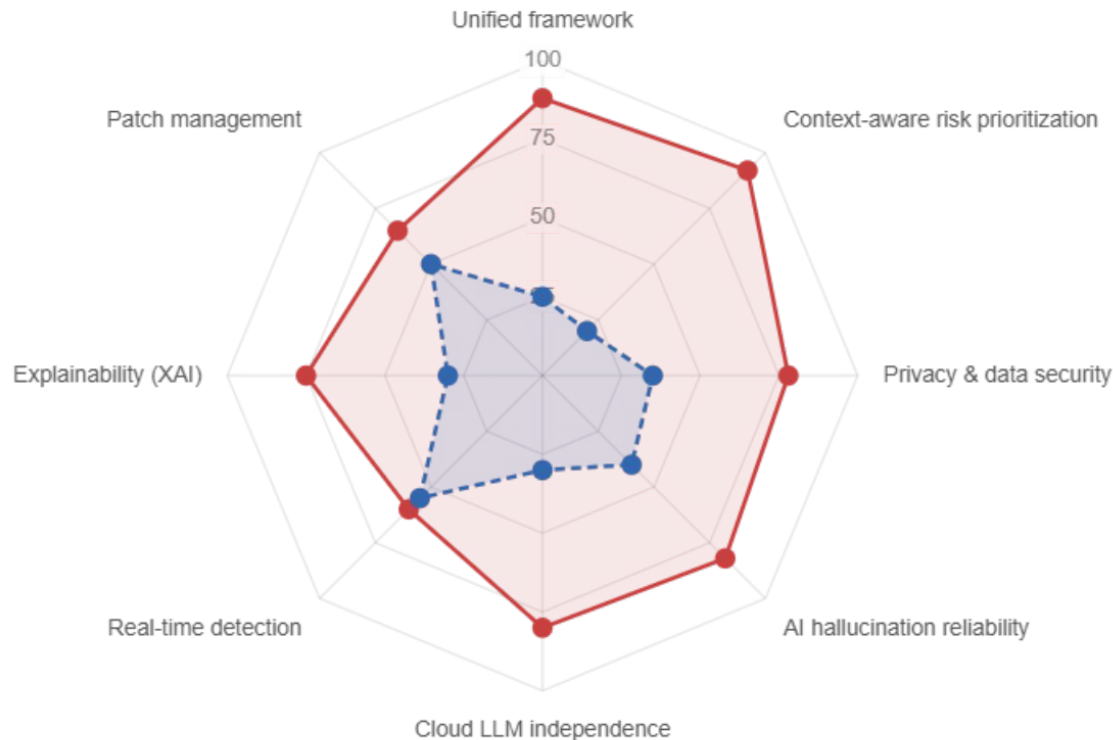
[10] BRACU Thesis (2024) — Tested AI-assisted security testing in a university setting with limited resources, showing these tools can work even in smaller organisations, not just big companies.

Research Gaps

- ☐ Heavy Dependence on Cloud-Based LLM Services
- ☐ Lack of Unified Vulnerability Assessment Framework
- ☐ Absence of Context-Aware Risk Prioritization
- ☐ Lack of Post remediation validation.
- ☐ Privacy and Data Security Concerns in Cloud-Based Systems

■ Coverage level in existing literature

■ Gap severity (research need)



Current Techniques & Limitations

Cloud-Based AI Tools (GPT, Gemini, etc.)

- ☐ High cost and cloud dependency
- ☐ Privacy and data security risks
- ☐ May produce incorrect outputs (hallucination risk)

Cloud-Based LLM Security Tools

- ❑ Cloud-based LLM security tools often rely on expensive external APIs such as GPT or Gemini.
- ❑ This increases operational costs and introduces privacy risks, as sensitive vulnerability data must be transmitted outside the local environment.

Independent Vulnerability Scanners (Nmap, ZAP, Nikto, OpenVAS)

- ❑ These tools operate separately without integration, leading to fragmented results and lack of a unified view of system vulnerabilities.

Basic Scan Report Systems

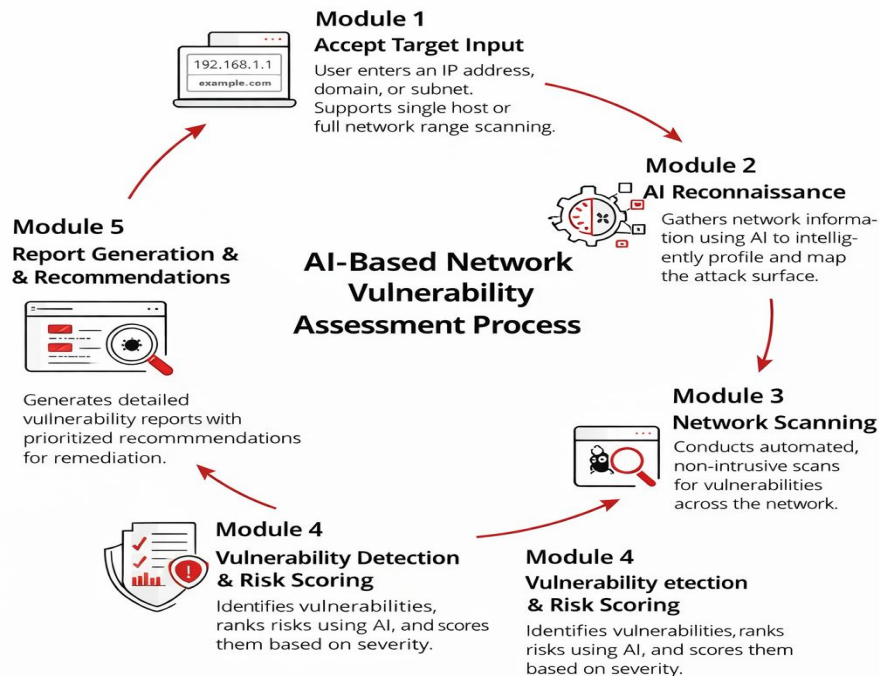
- ❑ They generate raw and unstructured outputs, often containing false positives and duplicate findings, with no intelligent prioritization of critical threats.

Controlled Environment Testing

- ❑ The framework was evaluated solely in a controlled academic setting, leaving its performance on live production systems, resource-constrained IoT edge devices, and low-latency real-time streams unvalidated.
- ❑ Compounding this, the authors themselves acknowledge that LLM integration, despite its demonstrated practical success, remains deferred to future work.

Methodology

- 1. Accept Target Input** User enters an IP address, domain, or subnet. Supports single host or full network range scanning.
- 2. Smart Scan Orchestrator** Automatically selects and launches the right tools based on the target — no manual setup needed. Tools: Nmap · Nikto · OWASP ZAP ↓ Reduces manual setup effort by 60–70%
- 3. Automated Safe Scanning** Runs comprehensive, non-destructive scans across the full attack surface.
 - 65,000+ ports & services — Nmap
 - 7,000+ web vulnerability checks — Nikto
 - OWASP Top 10 risks — ZAP
- 4. Unified Result Engine** Merges outputs from all tools into a single deduplicated finding store, eliminating overlapping detections. ↓ Reduces duplicate findings by 40–50%
- 5. AI Security Analyst** Local LLM via Ollama analyses consolidated findings and produces human-readable risk intelligence.
 - Prioritises risks by severity
 - Explains vulnerabilities in plain language
 - Filters out false positives ↓ Reduces false positives by 30–45%



6. CVE Intelligence Layer Cross-references every finding against official vulnerability databases for evidence-backed severity ratings.

- 250,000+ CVE records
- CVSS severity scores
- Official patch references

7. Smart Report Generator Produces a structured PDF or HTML report containing:

- Critical, Medium, and Low severity findings
- Severity rankings per finding
- AI-generated recommended fixes ↓ Reduces manual reporting time by 80%

8. Re-Scan Validation After fixes are applied, NAVAT re-runs targeted scans to confirm vulnerabilities are fully remediated and tracks security improvement over time.

- Confirms patch effectiveness
- Tracks security posture over time

References

- [1] S. Khan and S. Parkinson, "Review into state of the art of vulnerability assessment using artificial intelligence," in Guide to Vulnerability Analysis for Computer Networks and Systems, Springer, 2018, pp. 3–32.
- [2] M. Malkawi and R. Alhajj, "AI-powered vulnerability detection and patch management in cybersecurity: A systematic review," Mach. Learn. Knowl. Extr., vol. 8, no. 1, p. 19, 2026. DOI: 10.3390/make8010019
- [3] S. Chemparathy, N. Shivananjappa, and R. Creutzburg, "AI-based vulnerability scanners: A cross-sectional survey analysis," in Proc. IS&T Int. Symp. Electron. Imag., Art. no. MOBMU-326, 2025.
- [4] S. Chemparathy et al., "A holistic approach for cybersecurity vulnerability assessment based on open source tools Nikto, Acunetix, ZAP, Nessus, and ImmuniWeb," ResearchGate, 2024.
- [5] S. Chopra, H. Ahmad, D. Goel, and C. Szabo, "ChatNVD: Advancing cybersecurity vulnerability assessment with LLMs," arXiv:2412.04756, 2024.
- [6] B. Susilo, A. Muis, and R. F. Sari, "Intelligent intrusion detection system against various attacks based on a hybrid deep learning algorithm," Sensors, vol. 25, no. 2, p. 580, 2025. DOI: 10.3390/s25020580
- [7] R. Podder et al., "SPEAR: Security posture evaluation using AI planner-reasoning on attack-connectivity hypergraphs," arXiv:2506.01227, 2025.
- [8] G. Deng et al., "PentestGPT: An LLM-empowered automatic penetration testing tool," arXiv:2308.06782, 2023. DOI: 10.48550/arXiv.2308.06782
- [9] M. Patil, D. Thakare, A. Bhure, S. Kaundanyapure, and A. Mune, "An AI-based approach for automating penetration testing," IJRASET, vol. 12, pp. 5019–5028, Apr. 2024. DOI: 10.22214/ijraset.2024.61113
- [10] [Authors], "AI-based automated penetration testing," Undergrad. Thesis, Dept. CSE, BRAC Univ., 2024. [Online]. Available: <https://dspace.bracu.ac.bd/xmlui/handle/10361/27423>
- [11] Z. Yu et al., "ShotFlex: A reinforcement learning-based cyber attack path generation method," Security and Safety, vol. 4, Art. no. 2025003, 2025. DOI: 10.1051/sands/2025003
- [12] L. Yu, H. Ma, P. Xue, M. Hu, and H. Hao, "An efficient internet-wide port scan algorithm based on deep reinforcement learning," Future Internet, vol. 18, no. 3, p. 120, 2026. DOI: 10.3390/fi18030120